

Midterm 2 - Due by Monday 4/20

You are a logistics manager for a multinational oil company “Efficient Oil”. The company operates oil tankers that visit different ports offloading the load of the tanker until it has to return to the the company’s loading port. The loading port is numbered 0 and there are other $N = 15$ ports the company services. The company operates up to 5 tankers at a given time (quarterly cycles) each with capacity $C = 3$ million barrels. Each operating tanker $k = 1, \dots, K \leq 5$ is loaded with $q_k \leq C$ quantity of oil at port¹. Each travel of the tanker between two ports $i, j = 0, 1, \dots, N$ has a random traveling time $T_{i,j}$ in months that depends on the weather and other factors, historical observations of these traveling times are recorded in the file *sample_travel.times.csv*. The demand at each port D_i is also random and not known at the time the tankers depart. Depending on fluctuations of the oil prices and current needs of each port, historical demands are recorded in the file *hist_demands.csv*. An active tanker k with a tour of ports $s_k = (0, i, j, \dots, 0)$ where $s_k(0) = 0$, $s_k(1) = i$, $s_k(2) = j$ and so on. The tanker offloads a quantity $\tilde{q}_k(\ell) = (\tilde{q}_k(\ell - 1) - D_{s_k(\ell)})^+$ with $\tilde{q}_k(0) = q_k$, that is, the tanker is loaded to a level q_k starting its tour at port 0 and offloads $D_{s_k(\ell)}$ at each port if it has enough oil. The cost of operating a tanker is $v = \$100$ thousand per day of travel proportional to the load of the tanker $\tilde{q}_k(\ell)/C$ for a trip between $s_k(\ell), s_k(\ell + 1)$. That is, if for example the tanker k is initially loaded to level $q_k/C = 1/2$ and the first stop is port $s_k(1) = i$ with a traveling time taking $T_{0,s_k(1)}$ months, then the variable cost is $30T_{0,s_k(1)}v[\tilde{q}_k(0)/C]$ dollars. At every stop the price of oil is also variable $\tilde{p}$ and the revenue the tanker makes at each stop is $\min(D_{s_k(\ell)}, \tilde{q}_k(\ell))\tilde{p}$. In other words, the tanker can sell the minimum between its current load and the demand at a random price $\tilde{p}$.

Your model will run once per quarter and determine the number of active tankers (you can choose to operate less than the available tankers, and also cover less than all the N ports) and the load q_k of each of the active tankers. The model has to be implemented in a single script called *opt_tankers.py* with a function called *optimize_tankers* outputting a solution with the tours each tanker will perform $[s_1, s_2, s_3, s_4, s_5]$ and the quantities $[q_1, q_2, q_3, q_4, q_5]$ when operating all the 5 tankers. The workflow is as follows:

- You are given an integer denoting the upcoming optimization quarter 0, 1, 2, 3 and the number of tankers available $K \leq 5$.
- Your function *opt_tankers* outputs a list of tours $[s_1, \dots, s_K]$ and initial loading quantities $[q_1, \dots, q_K]$ that will be used for the whole quarter.
- Each tanker $k = 1, \dots, K$ travels their tour and comes back to the home tour taking a random amount of time $\sum_{i,j \in s_k} T_{i,j}$. If this time is more than 3 months, that tanker cannot be used in time for the next quarter.
- All tankers that arrive on time (less than 3 months) are the new K and the cycle starts again on the next quarter.

To ensure that the script works as intended you are given a unit test script *unit_tests.py* ensuring that the output of your model runs within the infrastructure of the company. For the same reason,

¹To make computation easier, q_k is rounded 1 decimal place.

you are also constrained that the script can only use packages contained in the file *requirements.txt*.

The performance of your strategy is evaluated over out-of-sample future quarters (you can assume that the quarters behave i.i.d. as in your sample data). For a given quarter, let $D_{i,t}, \tilde{p}_{i,t}$ be the random demand and price at port $j = 1, \dots, N$ for quarter t . Also let K_t be the number of active tankers at quarter t . The evaluation metric is defined as the out-of-sample operational profit (T is the number of future quarters):

$$\frac{1}{T} \sum_{t=1}^T \sum_{k=1}^{K_t} \sum_{\ell=1}^{|s_k|} [\tilde{p}_{s_k(\ell)} \min(D_{s_k(\ell),t}, \tilde{q}_k(\ell)) - 30vT_{s_k(\ell-1),s_k(\ell)}(\tilde{q}_k(\ell-1)/C)]$$

Task: Come up with an optimization model that performs best according to this metric.